
PHYSICAL AI ONCOLOGY TRIAL COMPETITION PROPOSAL

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Goal = End-to-End Trial Human Replacements: From 20 GenAI Papers to the Real World

Problem & Solution

- a) Problem: Oncology trials are manual, have data delays, coordinator burden, and workflows vary causing protocol deviations.
- b) Solution: Physical AI end-to-end platform (sensing + hybrid edge-cloud AI architecture + LLM tasks + human override, audits).
- c) Facilitates: Planning & design, site management, execution, monitoring & quality, regulatory compliance, data management.
- d) Tasks: 10/25 "End-to-End Oncology Clinical Trial LLM Efficiency For Industry Adoption with FDA/ICH Regulations." [1]
- e) Expansion: 1/26 Proposal now includes other human based tasks previously unobtainable by generative & agentic AI. [1]
- f) Progression: a) 1/6 Physics RL code modernized to 2026 [2, 3], b) 1/11 Demo: RL/Digital twin trial simulation upgrade. [4]

Stage I - Digital Body (Proposal)

Goal: Human replacement is trained to rival author regarding how to use generative AI in trial settings based on following tasks. [5]

- a) Image analysis [6, 7], Table analysis [8], Reverse-engineering [9], Multi-LLM problems [10], In-context table analysis. [11]
- b) Literature: 40 TIME papers [12], 100 Bevacizumab papers [13], 45 Drug cost papers [14], 20+ Industry solutions papers. [15]
- c) Kaplan-meier, forest, and violin plots [16], 10 yr trial data into meta-analysis [17], 40 meta-analyses into DT proposals. [4]
- d) 100K patient empirical code-free trial [18], QSP and Digital Twin trial simulations in Python. [19, 20, 21]
- e) ML drug synergy generation [22], Patient trial matching, AI peer review [23], LLM Python competition, FDA dataset. [24]
- f) 25+ LLM author papers to trial savings [1], Table code language conversions [25], 1500 lines of code generated. [26]

Stage II - Mobile Sensing Body

Goal: Human replacement senses and walks, maintaining prior intelligence training.

- a) The same human replacement brain gains "walk-and-know" for clinical site context via supervised mobility. [27]
- b) Head-to-head competition against other robots: Sensing/telepresence enables context-aware checklists, and room readiness. [28]
- c) Human-like robot is enabled for real-time task routing while maintaining full audit trails. [29, 30]

Stage III - Actuating Body

Goal: Final locomotion and manipulation policies run on an actionable prompt request and response receiving robot.

- a) The finished human replacement brain adds validated actuation for selected, low-liability physical workflows. [31, 32, 33]
- b) Multi-Robot tournaments [24]: Physical tasks for logistics and chain-of-custody checkpoints are enabled. [34, 35]
- c) The loop is now closed between decisions and physical execution. [36]

Utilized Works

- a) 1/26 Tang et al. "Deep Reinforcement Learning for Robotics: A Survey of Real-World Successes." [37]
- b) 12/25 Kinsley, H. Gemini 3 Pro generated robot code. "Training a Unitree G1 to Walk w/ Reinforcement Learning." [38]
- c) 4/25 Huhulea "Artificial Intelligence Advancements in Oncology: A Review of Current Trends and Future Directions." [39]
- d) 3/25 Sun, H. "Advancements in Clinical Trial Methodologies: A Century in Review." [40]
- e) 3/19 Sfera, D. "The Only Crash Course To Clinical Research You'll Ever Need (full 5 hour OFFICIAL video)." [41]
- f) 4/23 Catenacci, Daniel "27 Principles of Clinical Trials." [42]

Utilized RL Works

- a) 3/25 Amini, A. "MIT 6.S191: Reinforcement Learning, Introduction to Deep Learning." [43]
- b) 11/25 Python Simplified "Easiest Reinforcement Learning Explanation You'll Ever See!" [44]
- c) 8/24 IBM Technology "Reinforcement Learning from Human Feedback (RLHF) Explained." [45]
- d) 6/25 Computerphile "Reinforcement Learning, Reinforcement Learning is how robots test the water in the real world." [46]

Notes: Stage I Experimentation is a key requirement for generating onboard prompt requests; and understanding + taking action on offboard large language model responses. Example methods: Deterministic task graphs vs. State-based rules + heuristics vs. Small onboard agentic LLM. Latency exists for all offboard open-source or proprietary LLM responses.

References

- [1] Kevin Kawchak. End-to-end oncology clinical trial llm efficiency for industry adoption with fda/ich regulations, October 2025. doi:10.5281/zenodo.17451709.
- [2] Nicholas Renotte. Deep reinforcement learning tutorial for python in 20 minutes. URL: <https://www.youtube.com/watch?v=c05g5qLrLS0&list=PLgNJ02hghbmjlE6cuKMws2ejC54BTaaWV>.
- [3] Kevin Kawchak. URL: https://colab.research.google.com/drive/1m1KVxgkTUU1INBBtkhiW_94neWSpMGn5?usp=drive_link.
- [4] Kevin Kawchak. End-to-end pancreatic ductal adenocarcinoma digital twin clinical trial proposals, June 2025. doi:10.5281/zenodo.15735068.
- [5] Kevin Kawchak. Llm papers: Cancer drug & oncology trials, January 2026. doi:10.5281/zenodo.18171362.
- [6] Kevin Kawchak. Lmm chemical research with document retrieval. *ChemRxiv*, August 2024. doi:10.26434/chemrxiv-2024-p91gm.
- [7] Kevin Kawchak. Lmm spectrometric determination of an organic compound. *ChemRxiv*, August 2024. doi:10.26434/chemrxiv-2024-qtnkj.
- [8] Kevin Kawchak. High dimensional and complex spectrometric data analysis of an organic compound using large multimodal models and chained outputs. *ChemRxiv*, September 2024. doi:10.26434/chemrxiv-2024-06gf1.
- [9] Kevin Kawchak. Paclitaxel biosynthesis ai breakthrough. *ChemRxiv*, October 2024. doi:10.26434/chemrxiv-2024-pqjd3.
- [10] Kevin Kawchak. Monoclonal antibody bioprocess engineering advancements using conversational artificial intelligence. *ChemRxiv*, October 2024. doi:10.26434/chemrxiv-2024-3m7m1.
- [11] Kevin Kawchak. mab bioprocess engineering in-context table forecasts using conversational ai literature insight generations. *ChemRxiv*, December 2024. doi:10.26434/chemrxiv-2024-jzbj0.
- [12] Kevin Kawchak. Cancer vs. conversational artificial intelligence. *bioRxiv*, December 2024. doi:10.1101/2024.12.28.630597.
- [13] Kevin Kawchak. Clinical decision support based on bevacizumab cancer trials and pushing the limitations of advanced llms, January 2025. doi:10.5281/zenodo.14968162.
- [14] Kevin Kawchak. Cost containment of global monoclonal antibody drugs and cancer clinical trials via llm focused reasoning, February 2025. doi:10.5281/zenodo.14968404.
- [15] Kevin Kawchak. Autonomous llm agent and scalable reasoning llm for generating cancer drug industry cost solutions, March 2025. doi:10.5281/zenodo.15072843.
- [16] Kevin Kawchak. Ai revolution toward the cure of lung adenocarcinoma, April 2025. doi:10.5281/zenodo.15278152.
- [17] Kevin Kawchak. 10 year glioblastoma clinical trial meta-analyses by autonomous ai at scale. survival, hr, ae, and rob scored in ai reports and charts, including verifications, May 2025. doi:10.5281/zenodo.15549831.
- [18] Kevin Kawchak. Chatgpt 100,000 patient 24-month in silico phase iii 5-arm pancreatic cancer clinical trial triplicate, July 2025. doi:10.5281/zenodo.16415815.
- [19] Kevin Kawchak. Qsp metastatic pancreatic cancer ai clinical trial simulation from protocol to prediction: Code, vvuq, and playbook, August 2025. doi:10.5281/zenodo.17001137.
- [20] Kevin Kawchak. Accelerating fda compliance and cost efficiency of in silico clinical trials via ai digital twin pancreatic cancer simulation, September 2025. doi:10.5281/zenodo.17239510.
- [21] Kevin Kawchak. Daraxonrasib efficient llm trial simulations, December 2025. doi:10.5281/zenodo.18099351.
- [22] Kevin Kawchak. Llm-generated glioblastoma drug synergy machine learning: From rapid code prototypes to project deliverables package, November 2025. doi:10.5281/zenodo.17614396.
- [23] Kevin Kawchak. Ai peer review acceleration of llm-generated glioblastoma clinical trial patient matching ml, fda/ich/iso, and fastapi, November 2025. doi:10.5281/zenodo.17774560.
- [24] Kevin Kawchak. Code generation competition: 16 proprietary vs. open-source llms iterative learning based on fda adverse event reporting system, December 2025. doi:10.5281/zenodo.18029100.
- [25] Kevin Kawchak. Latex llm oncology table prompting guide, January 2026. doi:10.5281/zenodo.18113570.
- [26] Kevin Kawchak. Scaling oncology llm generated code: Jan 2026, December 2025. doi:10.5281/zenodo.18072433.
- [27] Lillian Hung et al. Reflexive case studies on conducting technology implementation research in long-term care homes. *Digital Health*, 11:20552076251354904, June 2025. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12205190/>, doi:10.1177/20552076251354904.

- [28] Lillian Hung, Joey Oi Yee Wong, Haopu (Lily) Ren, Yong Zhao, Jason Jincheng Fu, Jim Mann, and Lun Li. The impact of telepresence robots on family caregivers and residents in long-term care. *International Journal of Environmental Research and Public Health*, 22(5):713, May 2025. URL: <https://www.mdpi.com/1660-4601/22/5/713>, doi:10.3390/ijerph22050713.
- [29] Tetsuyou Watanabe, Jane Li, Maya Grace Torii, Yuka Miura, Misako Dai, Gojiro Nakagami, and Shinichi Hirai. Current state and future perspectives on robotic technology for improving efficiency of nursing care. *Advanced Robotics*, 39(23):1482–1505, December 2025. URL: <https://www.tandfonline.com/doi/full/10.1080/01691864.2025.2561630>, doi:10.1080/01691864.2025.2561630.
- [30] Feng Xiao, Youhai Zhang, Jie Fang, Xing Guo, and Rubing Huang. Improving the navigation optimization of hospital logistics robots under complex lighting changes by using improved orb-slam3 and deep learning visual slam algorithm. *Discover Applied Sciences*, 7(4):291, April 2025. doi:10.1007/s42452-025-06775-y.
- [31] Alejandra Rojas, Ismail Gölgeci, and Gabriele Santoro. Navigating meaningful work: Exploring the impact of mobile telepresence robots on healthcare professionals. *Technological Forecasting and Social Change*, August 2025. URL: <https://www.sciencedirect.com/science/article/pii/S0040162525001878>, doi:10.1016/j.techfore.2025.124156.
- [32] Yun Hsuan Lee, Fang Yu Hsu, and Angela Shin-Yu Lien. Health care professionals' perspectives of socially assistive robots in health care settings: Systematic review. *Journal of Medical Internet Research*, 27:e79634, October 2025. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12550450/>, doi:10.2196/79634.
- [33] Hyojin Shin, Seohee Kim, Eunseo Jung, Youjung Han, Nayoung Lee, Sangchan Park, Changhoon Jeon, and Jiyoung Woo. A comprehensive review of autonomous mobile robots in healthcare: Implications for patient-transporting human-following robots. *Robotics*, 14(12):179, November 2025. URL: <https://www.mdpi.com/2218-6581/14/12/179>, doi:10.3390/robotics14120179.
- [34] Namho Cho. Developing a checklist for implementing autonomous mobile robots (amrs) in healthcare facilities. *Research Gate*, November 2025. URL: https://www.researchgate.net/publication/397825377_Developing_a_Checklist_for_Implementing_Autonomous_Mobile_Robots_AMRs_in_Healthcare_Facilities.
- [35] Jiafeng Song et al. Clinicians' perceptions and potential applications of robotics for task automation in critical care: Qualitative study. *Journal of Medical Internet Research*, 27(1):e62957, March 2025. URL: <https://www.jmir.org/2025/1/e62957>, doi:10.2196/62957.
- [36] Jinkai Qiu, Yungjun Kim, Gaurav Sethia, Tanmay Agarwal, Siddharth Ghodasara, Zackory Erickson, and Jeffrey Ichnowski. Orb: Operating room bot, automating operating room logistics through mobile manipulation. *arXiv*, (arXiv:2509.15600), September 2025. arXiv:2509.15600. URL: <http://arxiv.org/abs/2509.15600>, doi:10.48550/arXiv.2509.15600.
- [37] Chen Tang, Ben Abbatematteo, Jiaheng Hu, Rohan Chandra, Roberto Martín-Martín, and Peter Stone. Deep reinforcement learning for robotics: A survey of real-world successes. *Annual Review of Control, Robotics, and Autonomous Systems*, 8(1):153–188, May 2025. URL: <https://www.annualreviews.org/content/journals/10.1146/annurev-control-030323-022510>, doi:10.1146/annurev-control-030323-022510.
- [38] Harrison Kinsley. Training a unitree g1 to walk w/ reinforcement learning. URL: <https://www.youtube.com/watch?v=FGnAeUXRZ4E>.
- [39] Ellen N. Huhulea, Lillian Huang, Shirley Eng, Bushra Sumawi, Audrey Huang, Esewi Aifuwa, Rahim Hirani, Raj K. Tiwari, and Mill Etienne. Artificial intelligence advancements in oncology: A review of current trends and future directions. *Biomedicine*, 13(4):951, April 2025. URL: <https://www.mdpi.com/2227-9059/13/4/951>, doi:10.3390/biomedicine13040951.
- [40] Haoxuan Sun. Advancements in clinical trial methodologies: A century in review. *International Journal of Applied Technology in Medical Sciences*, 4(1):10–17, March 2025. URL: <https://emiratesscholar.com/directory/index.php/ijatms/article/view/1037>, doi:10.54878/g50ysx32.
- [41] Dan Sfera. The only crash course to clinical research you'll ever need (full 5 hour official video). URL: <https://www.youtube.com/watch?v=yJNUq7bih8c&t=8276s>.
- [42] Daniel Catenacci. 27 principles of clinical trials. URL: https://www.youtube.com/watch?v=N-JK0gX5w4A&list=PL_m8UgCdT6MxkrE5w67oI4Q8jL2_5_ZCP&index=5.
- [43] Alexander Amini. Mit 6.s191: Reinforcement learning. URL: <https://www.youtube.com/watch?v=to-1HJfK4pw>.
- [44] Python Simplified. Easiest reinforcement learning explanation you'll ever see! URL: <https://www.youtube.com/watch?v=sgsQZmlJbSY>.
- [45] IBM Technology. Reinforcement learning from human feedback (rlhf) explained. URL: https://www.youtube.com/watch?v=T_X4XFwKX8k.
- [46] Computerphile. Reinforcement learning - computerphile. URL: https://www.youtube.com/watch?v=844U9T_S0rA.